Road Damage and Pothole Detection using YOLOv8

AI/ML Specialisation Capstone Project Problem Statement

Field	Details
Project Number	2
Domain	Smart Cities / Object Detection
Recommended Group Size	5 students
Primary Dataset	Roboflow Pothole Object Detection Dataset
Dataset Source URL	https://public.roboflow.com/object-detection/pothole
Core Curriculum Alignment	Deep Learning, TensorFlow/Keras, CNN/RNN/Transformers/Generative Models as applicable, Model Optimisation, Deployment, and MLOps.
Expected Prototype	Streamlit image upload app or FastAPI endpoint returning annotated images and JSON detections.

1. Background and Context

This capstone project is designed for a group of five learners to demonstrate end-to-end AI/ML capability. It connects model development with data preparation, performance evaluation, deployment readiness, and business interpretation. The project belongs to the Smart Cities / Object Detection domain and uses the Roboflow Pothole Object Detection Dataset dataset.

2. Problem Statement

City maintenance teams need a scalable way to identify potholes from road inspection images. Manual inspection is slow and inconsistent. The project requires learners to train an object detection model that localizes potholes in road images and produces bounding boxes with confidence scores.

3. Project Objective

Develop an object detection pipeline that detects potholes or road damage regions in images and provides model performance using object detection metrics.

4. Target Users and Use Case

Municipal road maintenance teams, smart-city analytics teams, road safety analysts, and infrastructure monitoring teams.

5. Dataset Details

Field	Details
Dataset Name	Roboflow Pothole Object Detection Dataset
Dataset Summary	The Roboflow public pothole dataset contains road images annotated with pothole bounding boxes and includes train-validation-test splits.
Dataset Access Type	Roboflow Public Dataset
Source URL	https://public.roboflow.com/object-detection/pothole
Recommended First Step	Download or load a small subset first, validate labels and file structure, then scale the model training pipeline.

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6. Steps to Access the Dataset

- Open the Roboflow public pothole dataset page.
- Create a free Roboflow account if export access is requested.
- Select the dataset version and export format as YOLOv8 or YOLOv5 PyTorch.
- Use the Roboflow-generated download code in the notebook, or download the ZIP and upload it to the working environment.
- Confirm that images and labels are organized into train, valid, and test folders.
- Inspect label files and bounding boxes before model training.
- If the dataset is too small, use additional Roboflow Universe pothole datasets, but document all sources clearly.

7. Scope of Work

In Scope

- Use a labelled pothole detection dataset with bounding boxes.
- Fine-tune a YOLOv8 model on pothole images.
- Evaluate detections using mAP, IoU, precision, recall, and qualitative error analysis.
- Create a demo app that accepts road images and returns annotated images.
- Package the inference pipeline for reproducible execution.

Out of Scope

- Real-time dashcam deployment on moving vehicles.
- Road severity scoring unless learners create a rule-based extension.
- Full GIS integration.

8. Recommended Technical Approach

- Define the problem clearly and convert it into a measurable machine learning task.
- Set up a reproducible project repository with folders for data access scripts, notebooks, trained models, reports, and deployment files.
- Perform dataset validation, exploratory analysis, preprocessing, and train-validation-test split creation.
- Build the recommended modelling pipeline: YOLOv8 nano/small as the main model, optional comparison with YOLOv5 or SSD..
- Track experiments with model configuration, validation metrics, training curves, and error observations.
- Tune key hyperparameters such as learning rate, batch size, number of epochs, optimizer, regularization, and model architecture choices.
- Evaluate on a holdout test set and prepare visual evidence such as confusion matrix, sample predictions, error cases, or mask/box overlays.
- Package the final model into a lightweight demo interface or API and document how to run it.

9. Model Evaluation Plan

Primary evaluation metrics: mAP@0.5, mAP@0.5:0.95, precision, recall, IoU, false positive/false negative examples, inference time per image.

The group should not report only one headline metric. The evaluation must include overall performance, class-wise or segment-wise performance where relevant, failure cases, latency, and a brief explanation of whether the model is usable for the stated user context.

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10. Deployment Expectation

Streamlit image upload app or FastAPI endpoint returning annotated images and JSON detections.

- The final prototype should load the trained model artifact without requiring retraining.
- The app/API should accept a user input in the expected format and return model output in a clear format.
- The demo should include at least three sample predictions and one known failure case.
- The README should include setup steps, dependencies, and run instructions.

11. Suggested Team Structure for 5 Students

Role	Responsibility
Student 1 - Project Lead & Business Analyst	Owns problem framing, stakeholder context, scope control, final narrative, and presentation coordination.
Student 2 - Data Engineer	Owns dataset access, data validation, preprocessing pipeline, train-validation-test splits, and reproducibility.
Student 3 - Modeling Lead	Owns baseline model, advanced model implementation, training experiments, and tuning strategy.
Student 4 - Evaluation & Research Lead	Owns metrics, error analysis, fairness/ethics checks, benchmarking, and limitation documentation.
Student 5 - Deployment & Documentation Lead	Owns Streamlit/FastAPI demo, Docker or deployment packaging, user guide, and final report formatting.

12. Expected Deliverables

- Problem statement and scope document.
- Data access and preprocessing notebook/script.
- EDA notebook with key observations and data-quality checks.
- Baseline model notebook and advanced model notebook.
- Model evaluation report with metrics, plots, and sample errors.
- Deployment prototype using Streamlit, FastAPI, or equivalent.
- Final presentation deck with problem, method, results, limitations, and future scope.
- GitHub/project folder with README and reproducible instructions.

13. Success Criteria

- Dataset is accessed from a documented public source and loaded reproducibly.
- The group demonstrates a meaningful baseline and an improved advanced model.
- Evaluation metrics are appropriate for the task and interpreted correctly.
- The deployed prototype works on new sample inputs.
- The final explanation clearly connects technical results to user/business value.
- Limitations, risks, and responsible use considerations are stated honestly.

14. Key Risks and Mitigation

Risk	Suggested Mitigation
Small dataset size may lead to overfitting.	Validate assumptions early, document limitations, and use error analysis to guide improvements.
Lighting, camera angle, and road texture variations can reduce generalization.	Validate assumptions early, document limitations, and use error analysis to guide improvements.
Bounding box quality directly affects model performance.	Validate assumptions early, document limitations, and use error analysis to guide improvements.

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Responsible use concern	The system should assist inspection prioritization and should not replace physical verification of road hazards.

15. Final Presentation Guidance

- Slide 1: Project title, team members, and problem context.
- Slide 2: Dataset source, access method, data size, and sample records/images.
- Slide 3: EDA and preprocessing decisions.
- Slide 4: Baseline model and advanced model architecture.
- Slide 5: Experiment tracking and hyperparameter choices.
- Slide 6: Evaluation results and error analysis.
- Slide 7: Deployment demo and inference workflow.
- Slide 8: Risks, limitations, ethical considerations, and future enhancements.

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